State Machines – Part 5
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Review

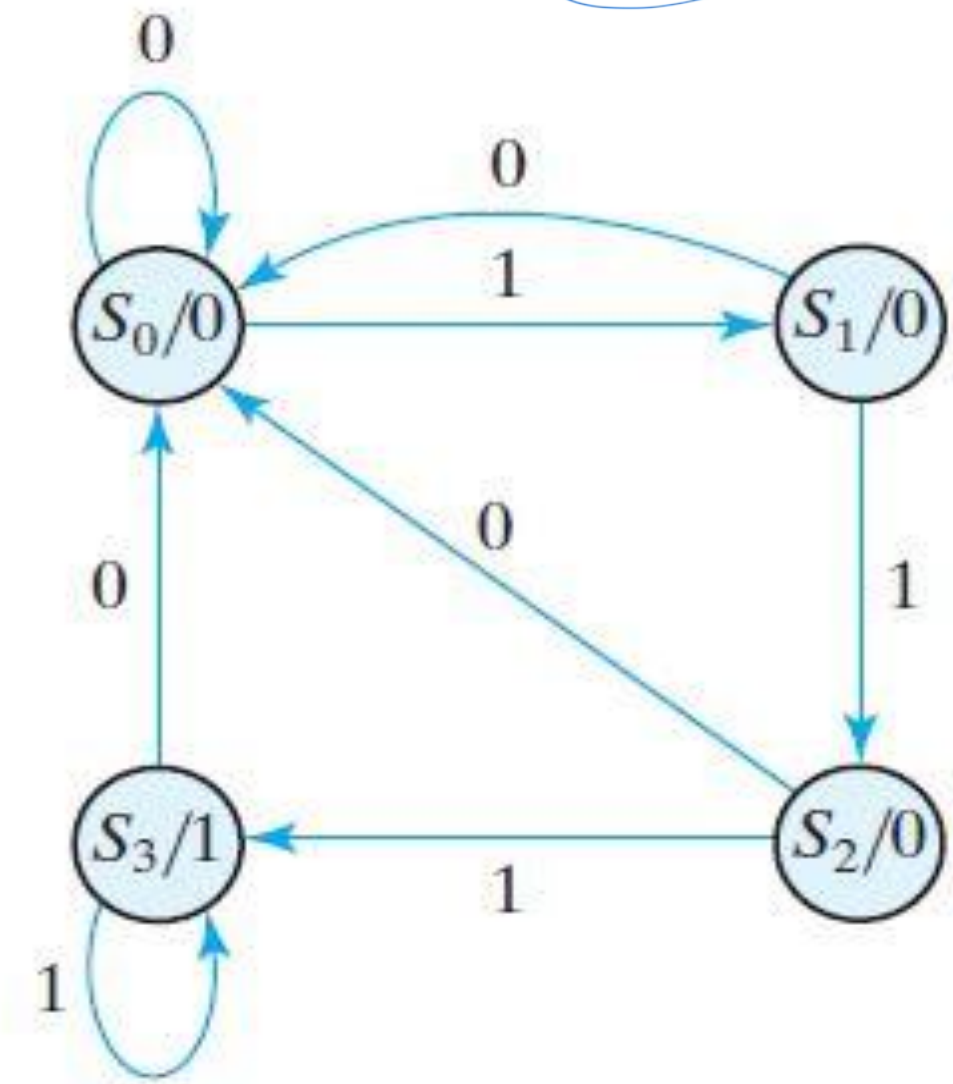
- We have discussed terminology relating to state assignment.
- We have also outlined the design procedure.

Design of a sequence detector

- We indicated that the goal is to design a sequence detector i.e., a circuit that detects a specific sequence of bits in the input stream.
- Let's assume that, in particular, we want a circuit that detects a sequence of three or more consecutive 1's in the incoming stream of bits. The state diagram for this would be like the one shown next.

State Diagram

Moore



Explanation

- S_0 is the reset (start) state. If the input is 0, the circuit stays in S_0 but if the input is 1, it goes to state S_1 to indicate that a 1 was detected.
- If the next input is 1, the circuit moves to state S_2 which shows that two consecutive 1's have arrived. If the input is 0, the circuit goes back to state S_0 .

More on the state diagram

- If we get a third consecutive 1, the circuit will move to state S_3 . A 0 input at this stage will send the circuit to state S_0 .
- The circuit will remain in state S_3 if more 1's appear in the input.
- The outputs are also indicated inside the circles that have the present states. Note that the output is 0 unless we have reached S_3 since three consecutive 1's in input would be necessary to get an output of 1.

Mealy or Moore model here ?

- Since the output is 1 when the circuit is in state S_3 and 0 otherwise, we have a Moore model sequential circuit here.
- Recall that, in the Moore model, the output depends only on the present state. Here our goal is to output 1 (or success) if we have got three (or more consecutive 1's) and this happens only if we have reached S_3

Next step ..

- We need to assign binary codes to the states and construct the state table.
- We will assume classical binary coding is adopted for state assignment.
- Therefore, we will have 2 flip-flops (and codes of 00, 01, 10 and 11 for states S_0 , S_1 , S_2 and S_3 respectively). For now, we need not worry about the type of flip-flop.

State Table

Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

S_0 S_1 S_2 S_3

Explanation

- The state table has labels A and B which correspond to the output of flip-flops.
- There are four states, given by 00, 01, 10 and 11 and at each of these states, the input (x) can be 0 or 1. This explains the first three columns (from the left) in the table.
- The entries in the next state columns depend on the input x as well as the present state. For example, when x = 1 and we are in state S_1 (i.e., code of 01), the next state is S_2 (i.e., code of 10).

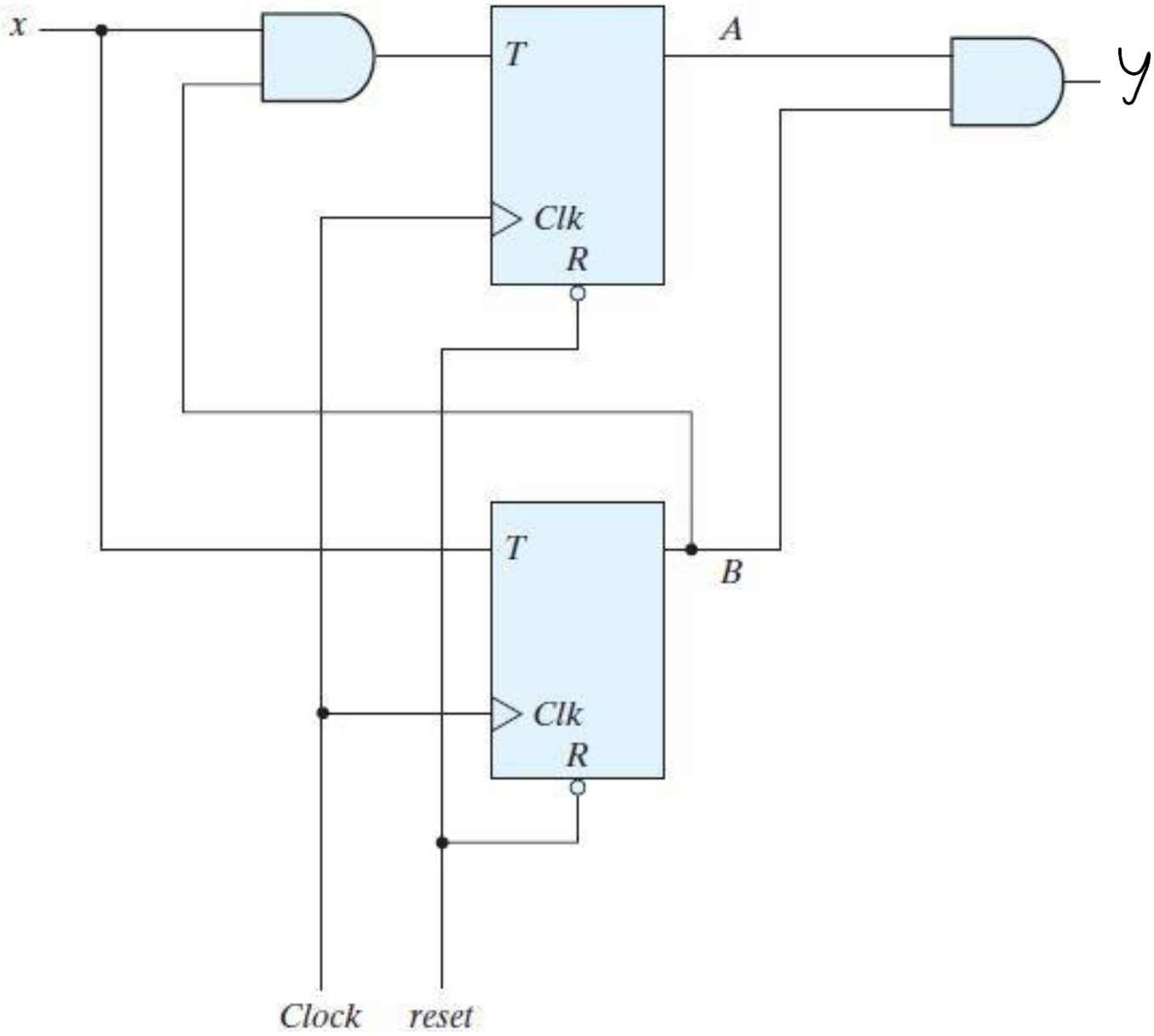
More on the state table

- Also, when we are in state S_2 (i.e., code 10) and the input (x) is 1, the next state will be S_3 (i.e., code 11).
- The output (y) will be 1 only when the present state is S_3 (i.e., code 11). That is, we have reached S_3 from S_2. At this stage (when we are in S_3), even if the input (x) is 0, it will only affect the entries in the columns corresponding to the next state.
- We now need to see what the inputs to the two flip-flops should be (and draw the circuit).

What are we exactly trying to do ?

- Our sequence detector will be a circuit which looks somewhat like the sequential circuits we saw in earlier parts (especially, the Moore versions).

Sample Circuit



Let's examine the sample circuit a little

- The sample sequential circuit has a pair of flip-flops. We know that the outputs for our sequence detector are A and B just like the flip-flop outputs of this sample circuit.
- However, it is not enough to know the outputs of the flip-flops alone. We need to know (i) what the inputs should be for the flip-flops and (ii) what the circuit should be for the output (y).
- Finding the inputs for the flip-flops is easy when we have D flip-flops since the characteristic equation for a D flip-flop is $Q(t+1) = D_Q$

On the characteristic equation

- The characteristic equation of a D flip-flop (just discussed) shows a direct relationship between the next state and the D input.
- It indicates that whatever is at D is showing up at the output of the flip-flop when there is a clock signal. So the flip-flop input equations can be obtained from the next-state columns of A and B in the case of D flip-flops. So we will choose D flip-flops for our design.

The equations when we use D flip-flops

$$A(t+1) = D_A(A, B, x) = \Sigma(3, 5, 7)$$

$$B(t+1) = D_B(A, B, x) = \Sigma(1, 5, 7)$$

$$y(A, B, x) = \Sigma(6, 7)$$

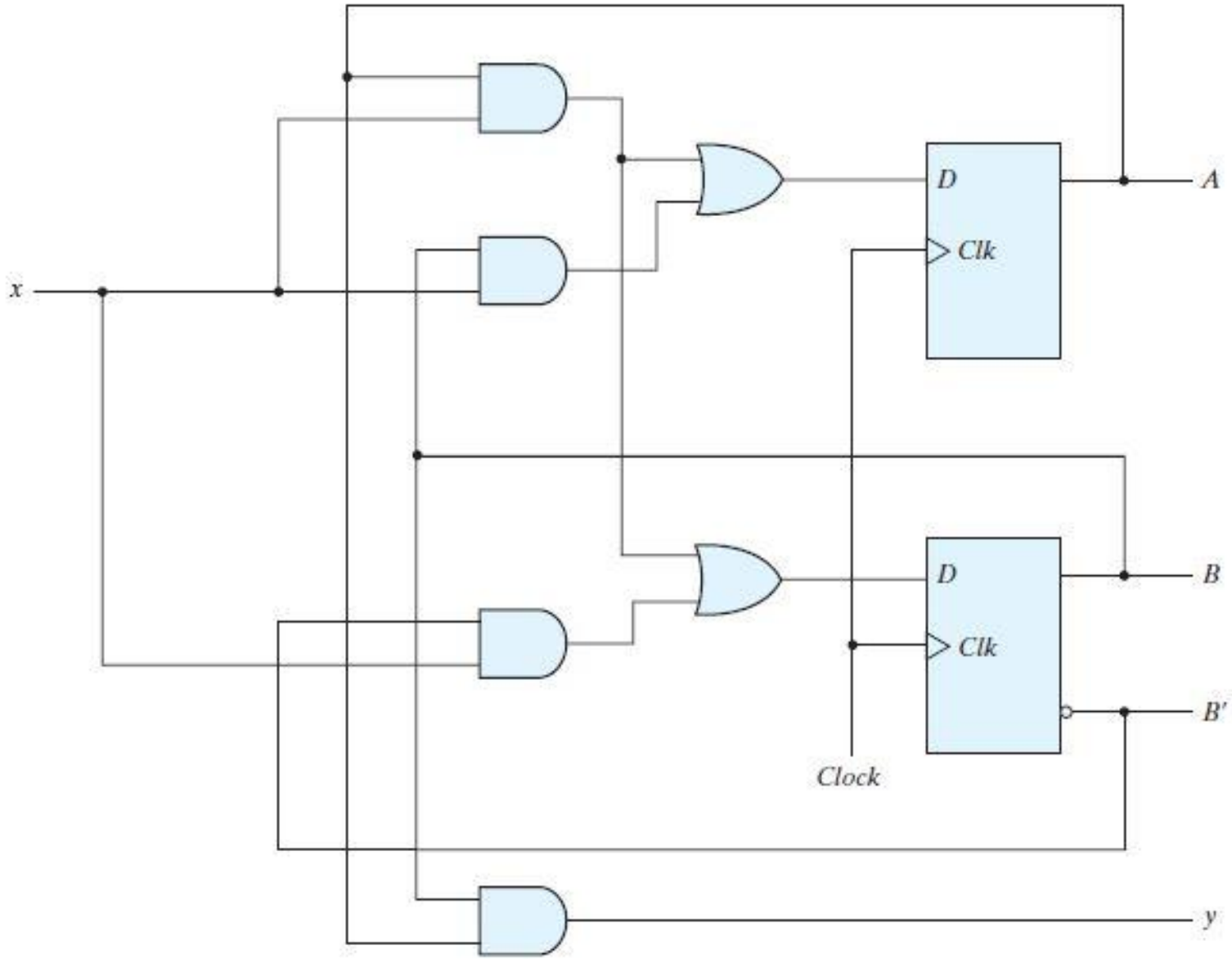
D_A and D_B above are input equations

Simplified Equations

- We can simplify the equations using K-maps and obtain

$$D_A = Ax + Bx$$
$$D_B = Ax + B'x$$
$$y = AB$$

The sequence detector circuit



What about other flip-flop types ?

- The input equations cannot be directly derived from the state table in the case of JK flip-flops or T flip-flops.
- We need to do some extra work. We will describe the design of a 3-bit binary counter via T flip-flops in the next part.

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- C. Roth and L.L. Kinney, Fundamentals of Logic Design, Gamma Publications, 2021.